



97770

BCA - NEP 4 Year 4th Semester

Examination, May-2026

COMPUTER GRAPHICS

Paper : 24BCA404DS04

Time allowed : 3 Hours]

[Maximum marks : 50

30+

Note : Question 1 is compulsory having five parts and each part is of 2 marks total of 10 marks and the remaining questions are of 10 marks. And attempt one question from each section.

1. Write a short note on the following : $5 \times 2 = 10$ 18
- (a) Define pixel.
 - (b) List the steps in scan conversion.
 - (c) What is the use of transformation matrices?
 - (d) What is a clipping window?
 - (e) Define spline and its importance.

Unit-I

2. Discuss interactive and non-interactive graphics with applications. 10 19
3. Explain video display devices and input devices in graphics systems. 10

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(2)
Unit-II

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4. Implement Bresenham's line algorithm for a line from (2, 3) to (10, 8). 10 / 8

5. Describe mid-point circle drawing algorithm. 10

Unit-III

6. Prove that two successive rotations are additive in 2D transformation. 10

7. Explain 2D reflection and shearing with example matrices. 10 / 3

Unit-IV

8. Write a short note on Polygon clipping. 10 / 10

9. Differentiate between viewport and window. Explain mapping between them. 10

$$\begin{array}{r} 16 \overline{) 674.1875} \\ -64 \\ \hline 30 \\ -16 \\ \hline 140 \\ -128 \\ \hline 120 \\ -112 \\ \hline 80 \\ -80 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 16 \\ \times 8 \\ \hline 128 \\ 16 \times 7 \\ \hline 112 \end{array}$$

$$\begin{array}{r} 25 \overline{) 3375} \\ -25 \\ \hline 87 \\ -75 \\ \hline 125 \end{array}$$

$$\begin{array}{r} 32 \\ \times 7 \\ \hline 224 \\ 256 \\ \hline 97770 \end{array}$$

$$\begin{array}{r} 120 \\ -112 \\ \hline 80 \\ -80 \\ \hline 0 \end{array}$$

$$\begin{array}{r} 0.281 \\ 32 \overline{) 90} \\ -64 \\ \hline 260 \\ -256 \\ \hline 40 \\ -32 \\ \hline 8 \end{array}$$

$$\begin{array}{r} 0.375 \\ 8 \overline{) 301} \\ -24 \\ \hline 60 \\ -56 \\ \hline 40 \end{array}$$

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